

ROHITH K

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Education

CGPA: 7.6/10	B.E. CSE (AI&ML) , Sri Eshwar College of Engineering	2024 – 2028
Percentage: 81%	Higher Secondary (Class XII) , Sri Chaitanya Techno School	2023 – 2024
Percentage: 90.5%	Secondary (Class X) , Sri Chaitanya Techno School	2021 – 2022

Experience

MERN – Stack Intern – *Better Tomorrow* 03 Dec 2025 – 21 Dec 2025

- Identified critical limitations in existing pharmacy platforms: no role-based access, unsecured endpoints, and manual checkout flows; constructed a production-grade MVC solution (React, Node.js, MongoDB) with JWT-authenticated REST APIs and AWS (EC2, S3), eliminating these gaps and achieving zero checkout failures with 99%+ API uptime.

Tech Stack: React, Node.js, Express.js, MongoDB, JWT, REST APIs, AWS (EC2, S3)

Projects

MITRA – Multilingual Voice AI Agent for Government Scheme Discovery Feb 2026 | [Link](#)

- Overcame critical Lambda cold-start latency and stateless session limitations in existing voice AI prototypes; architected a **zero - cost serverless pipeline** (AWS Lambda, DynamoDB, Amplify, Qdrant Vector DB) with Docker containerisation, resolving state persistence and reducing inference overhead to **near-zero** through vector-optimised semantic retrieval.

Tech Stack: Python, AWS Lambda, API Gateway, DynamoDB, React, Sarvam AI API (ASR/TTS), Docker, Qdrant

AquaPlastix – Polarized Imaging Microplastic Analyzer Feb 2026 | [Link](#)

- Innovated an end-to-end optical sensing setup – custom polarised light source, cross-polarised camera configuration, and controlled environment – deployed on Raspberry Pi with a **PyTorch & TensorFlow-trained classification model**, improving validation accuracy from **89% to 92%** through targeted data augmentation techniques, validated through a filed patent.

Tech Stack: Python, Tensorflow, ResNet-50, Raspberry Pi, Roboflow

Farm2Bag – Health Aware Grocery Recommendation System Mar 2025 | [Link](#)

- Addressed cold-start and relevance degradation in grocery recommendation by engineering a **multi-algorithm hybrid pipeline** – **K - Means (K=3)** demographic segmentation, **Apriori/FP - Growth** association rule mining, collaborative filtering, and **NLP** – driven medical condition extraction – converging behavioural, contextual, and domain intelligence into a unified inference layer.

Tech Stack: Python, Pandas, NumPy, Scikit-learn, MLxtend, spaCy, Flask, JavaScript, HTML/CSS, Vector DB

Skills

Languages	Python · JavaScript · Java · C · C++
Frameworks & Libraries	PyTorch · TensorFlow · Scikit-learn · ResNet-50 · Pandas · NumPy · React · Express.js
AI & NLP	LLM APIs · RAG · Prompt Engineering · ASR · Vector Embeddings
Data & Visualization	Pandas · NumPy · Matplotlib · Seaborn
Databases	MongoDB · MySQL · Qdrant · DynamoDB
Cloud & Tools	AWS (Lambda · EC2 · S3 · Amplify · API Gateway) · Docker · Roboflow · Git · GitHub

Coding Profiles

LeetCode : solved 100+ problems | **Skillrack :** solved 300+ problems

Certifications

Udemy – Data Structures & Algorithms using C and C++	2026
Anthropic – AI Fluency for Students	2026
Tata Group – GenAI Powered Data Analytics Job	2025
Pumo Technovation – Certified Full Stack Python Developer	2025

Achievements

AI Conclave 2025	Won INR 10,000 cash prize for Farm2Bag at Kongu Engineering College
PALS Innawah 2026	National Cluster Finalist among 300+ teams and recognised by academic and industry judges for innovation in edge AI and optical sensing
AI for Bharat Hackathon	Selected among 50 teams nationwide to advance to the project implementation phase